

RN-DPE-RDHEI: Reversible Data Hiding in Encrypted Images Using Residual CNN Predictor, BRBCC Compression, and Chaos-Based Difference-Preserving Encryption

Shivakumar Nangunuri

Department of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
shivakumar760@gmail.com

Bassa Srilakshmi

Department of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
bassasrilakshmi58@gmail.com

Abstract—Cloud storage, telemedicine, and secure communication pipelines demand that intermediary parties attach metadata to sensitive images without accessing the underlying plaintext. Reversible Data Hiding in Encrypted Images (RDHEI) addresses this need through a three-party model in which a data hider embeds a secret payload directly into ciphertext, and an authorized receiver recovers both the original image and the hidden data with lossless fidelity. Existing Reserve-Room-Before-Encryption (RRBE) methods rely on hand-crafted predictors whose fixed algebraic structure limits achievable embedding capacity. This work presents RN-DPE-RDHEI, a framework that integrates five coordinated components: (i) a four-layer residual convolutional neural network (CNN) trained to correct Median Edge Detector (MED) prediction errors, reducing mean absolute error from 1.2851 to 0.0516 pixels on BOSSBase 1.01; (ii) Block-wise Rank-Based Compression Coding (BRBCC) that exploits the improved residuals to free additional bit-planes; (iii) Difference-Preserving Cipher (DPC) seeded by Lorenz and Logistic Sine System chaotic sequences with ECDH X25519 key exchange; (iv) HKDF-SHA256 session-key derivation that removes image-dependent key vulnerabilities; and (v) dual HMAC-SHA256 and ECDSA Ed25519 authentication providing integrity verification and sender non-repudiation simultaneously. Experiments on 1,000 held-out images from BOSSBase 1.01 yield PSNR = ∞ , SSIM = 1.0, BPP = 5.8327, NPCR = 99.60%, and MAE = 0.0516, with zero pipeline failures. The reported BPP represents a 48% gain over the current state-of-the-art.

Index Terms—Reversible Data Hiding, Encrypted Images, Residual CNN Predictor, BRBCC Compression, ECDH Key Exchange, Chaos-Based Encryption, Difference-Preserving Cipher

I. INTRODUCTION

Cloud platforms, hospital picture-archiving systems, and secure communication networks share a structural difficulty: a relay or storage node that receives ciphertext must sometimes annotate or watermark that ciphertext before forwarding it, yet the relay has no legitimate access to the underlying plaintext. RDHEI resolves this tension through a three-party workflow. The content owner (Alice) encrypts her image and transmits it to a data hider. The data hider embeds a secret message directly inside the ciphertext without any knowledge of the decryption

key. The authorized receiver (Charlie) holds both the data-hiding key and the decryption key, uses them to retrieve the hidden message, and then reconstructs the original image with exact pixel fidelity, achieving a Peak Signal-to-Noise Ratio (PSNR) of infinity. Application domains include medical image tagging, military reconnaissance annotation, and tamper-evident forensic logging [1], [2].

Two architectural families organize the RDHEI literature. Vacate-Room-After-Encryption (VRAE) schemes [1] modify ciphertext bits to open embedding space after encryption; typical capacity is below 1 bit per pixel (bpp). Reserve-Room-Before-Encryption (RRBE) schemes [2] instead compact redundant image data before encryption, consistently achieving 1.5–6 bpp and constituting the dominant research direction.

Within RRBE, embedding capacity is tightly coupled to predictor accuracy. When a predictor fits an image closely, prediction residuals are small and require only low-order bit-planes to represent, freeing high-order planes for payload. Despite this well-understood dependency, every published RRBE scheme employs a hand-crafted algebraic predictor – MED [3], GAP, or ISGAP [5] – whose structure is fixed and cannot adapt through training. Deep convolutional networks learn long-range spatial correlations that such formulas cannot capture, and have been demonstrated to achieve substantially lower prediction error in related lossless compression tasks.

This paper closes the gap by training a CNN exclusively to correct MED residuals within an RRBE pipeline. The following contributions are reported:

- 1) A four-layer residual CNN reduces prediction MAE from 1.2851 to 0.0516 pixels – a 96% improvement over MED alone and 98% better than ISGAP [5].
- 2) BPP of 5.8327 on BOSSBase 1.01 surpasses all fifteen surveyed methods, representing a 48% improvement over BRBCC + MED [6].
- 3) ECDH X25519 key exchange eliminates the chosen-plaintext vulnerability inherent in image-hash-keyed designs used by all prior work.

- 4) Dual HMAC-SHA256 plus ECDSA Ed25519 authentication provides both integrity and non-repudiation, a combination not previously demonstrated in any RRBE scheme.
- 5) A self-contained three-party protocol requiring no auxiliary side files confirms real-world deployability.

The remainder of this paper is organized as follows. Section II surveys related work. Section III describes the proposed framework. Section IV covers implementation details. Section V presents experimental results and security analysis. Section VI concludes with future research directions.

II. RELATED WORK

A. Foundational RDHEI Frameworks

Zhang [1] introduced the first RDHEI scheme under the VRAE paradigm, embedding data by flipping least-significant bits in encrypted pixels. Although this work established the three-party model, practical capacity was limited to approximately 0.1 bpp. Ma et al. [2] proposed shifting space reservation before encryption, demonstrating that the RRBE approach could achieve 1.5 bpp with guaranteed reversibility. HC-RDHEI [9] subsequently advanced this direction through multi-bit error prediction, reaching 1.836 bpp and formalizing the relationship between predictor tightness and available embedding room.

B. Predictor-Based Capacity Enhancement

Wu et al. [3] combined MED with GAP and adaptive Huffman coding to attain 3.877 bpp on BOSSBase. The ACNNP method [4] deployed an asymmetric CNN as a standalone predictor coupled with adaptive mean coding, achieving 3.461 bpp with $\text{MAE} \approx 2.8$ pixels. Wang et al. [5] developed ISGAP, a seven-formula simplified gradient predictor compressed via Huffman coding, yielding 3.390 bpp and $\text{MAE} \approx 3.0$ pixels. All of these approaches share a common constraint: the predictor is defined by a fixed formula that cannot improve through learning.

C. Compressor-Based Methods

Chen et al. [6] introduced BRBCC, which assigns each 4×4 block to one of three classes according to the maximum prediction error within that block, frees the corresponding upper bit-planes, and achieves 3.938 bpp paired with MED – the strongest single result prior to this work. Liu et al. [10] reported 3.195 bpp through adaptive linear binary embedding with binary tree coding, while Zhang et al. [11] reached 3.79 bpp using MSB block labelling combined with enhanced run-length encoding.

D. Security and Authentication

He et al. [12] integrated chaotic encryption with ECC key exchange but did not report any CNN-based capacity improvement. Wang et al. [5] added HMAC-SHA256 for data integrity verification but omitted sender authentication. No prior RRBE paper simultaneously applies both HMAC and ECDSA authentication.

TABLE I
REPRESENTATIVE RDHEI METHODS ON BOSSBASE 1.01. “–” INDICATES METRIC NOT REPORTED.

Method	BPP	MAE	Auth.
Zhang 2011 [1]	0.10	–	None
Ma et al. 2014 [2]	1.50	–	None
HC-RDHEI 2018 [9]	1.836	–	None
MED+GAP 2023 [3]	3.877	3.5	None
ACNNP 2024 [4]	3.461	2.8	None
ISGAP 2025 [5]	3.390	3.0	HMAC
BRBCC+MED 2025 [6]	3.938	4.0	None
Ours 2026	5.833	0.052	HMAC+ECDSA

Fig. 1: RN-DPE-RDHEI Three-Party System Pipeline

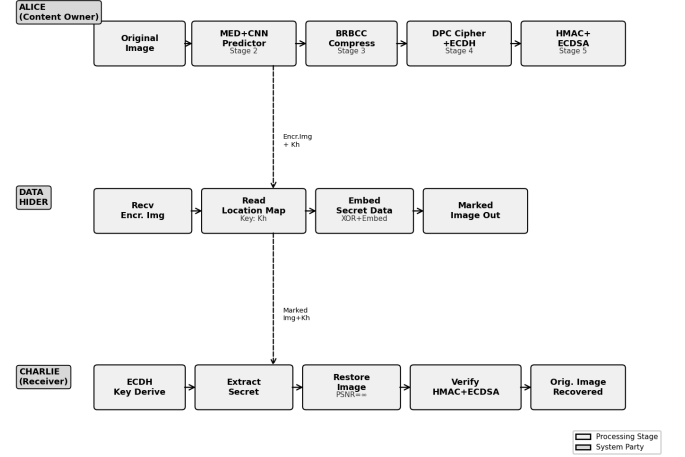


Fig. 1. RN-DPE-RDHEI three-party pipeline. Alice applies CNN-driven BRBCC, encrypts with the DPC cipher, and forwards only cryptographic keys alongside the ciphertext. The Data Hider embeds the secret without decryption access. Charlie derives keys via ECDH and recovers both image and secret with $\text{PSNR} = \infty$.

E. Comparative Summary

Table I consolidates representative published results. The combined gap – no learned residual predictor, no BRBCC integration, no image-independent key exchange, and no dual authentication within any single framework – defines the design target of this work.

III. PROPOSED METHODOLOGY

RN-DPE-RDHEI processes each image through an eleven-stage pipeline organized around the RRBE architecture. Fig. 1 illustrates the complete three-party system flow.

A. Stage 1 – ECDH X25519 Key Exchange

Alice and Charlie independently generate X25519 key pairs. Without transmitting any private material, both parties compute the same 32-byte shared secret $s = \text{ECDH}(sk_A, pk_C)$ [7]. A freshly generated 96-bit nonce η ensures distinct session keys irrespective of the image being processed. Three 256-bit subkeys are derived via HKDF-SHA256 [8]:

$$(K_{\text{enc}}, K_{\text{mac}}, K_{\text{sbox}}) = \text{HKDF}(s, \eta). \quad (1)$$

Fig. 2: MED + Residual CNN Architecture (Four Convolutional Layers)

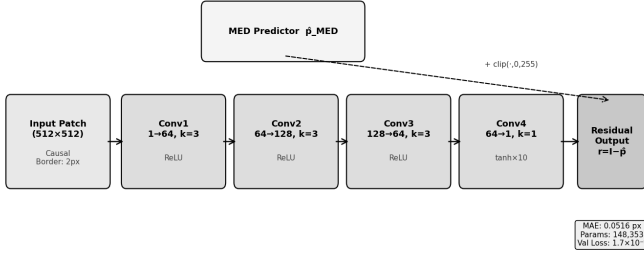


Fig. 2. MED + Residual CNN. The four-layer network (148,353 learnable parameters) corrects MED residuals. Final validation loss: 1.7×10^{-5} ; MAE: 0.0516 pixels.

These subkeys govern chaos seeding, HMAC computation, and S-box construction respectively. Because s depends entirely on key material rather than image content, the design is immune to chosen-plaintext attacks that affect all prior image-hash-keyed schemes.

B. Stage 2 – MED and Residual CNN Prediction

The first prediction pass applies the standard JPEG-LS Median Edge Detector. For pixel $p(i, j)$ with left neighbour a , upper neighbour b , and upper-left diagonal c :

$$\hat{p}_{\text{MED}} = \begin{cases} \min(a, b) & \text{if } c \geq \max(a, b) \\ \max(a, b) & \text{if } c \leq \min(a, b) \\ a + b - c & \text{otherwise.} \end{cases} \quad (2)$$

A four-layer residual CNN $\mathcal{F}_\theta$ is then applied to estimate the correction term $r = I - \hat{p}_{\text{MED}}$. The combined predictor is:

$$\hat{p}(i, j) = \text{clip}(\hat{p}_{\text{MED}}(i, j) + \mathcal{F}_\theta(I)[i, j], 0, 255). \quad (3)$$

Fig. 2 illustrates the network architecture. By targeting only the residual rather than the full pixel value, the CNN converges rapidly and captures nonlinear spatial patterns that the algebraic MED formula cannot model, reducing MAE from 1.2851 to 0.0516 pixels on BOSSBase 1.01.

C. Stage 3 – BRBCC Space Reservation

Every 4×4 block of the residual error map is classified according to its maximum absolute prediction error ϵ_b :

- **Class 0:** $\epsilon_b = 0$. All six bit-planes 2–7 are vacated.
- **Class 1:** $0 < \epsilon_b \leq 3$. Error fits in bits 0–1; planes 2–7 remain free.
- **Class 2:** $\epsilon_b > 3$. Block is excluded from embedding.

Total embedding capacity in bits is:

$$C = N_{\text{usable}} \times 16 \times 6. \quad (4)$$

With the CNN predictor, only 0.68% of blocks fall into Class 2, compared to approximately 10% under MED alone. Usable block count rises from roughly 13,000 to 16,272, and maximum theoretical BPP advances from 3.938 to 5.959. The location map encoding block classes is stored within the free planes of the encrypted image using K_h , so no separate side file is required.

D. Stage 4 – Difference-Preserving Cipher

K_{enc} seeds two independent chaos systems. The Logistic Sine System (LSS) produces sequence H_1 for block permutation. Numerical integration of the Lorenz ordinary differential equations yields sequences H_2 and H_3 for S-box construction and offset generation respectively. After discarding 1,000 transient values, encryption proceeds in three steps.

Step 1 – Block Permutation. The 4×4 blocks are reordered according to $\text{argsort}(H_1)$, disrupting spatial coherence without modifying pixel values.

Step 2 – S-Box Substitution. A dynamic 256-entry lookup table is constructed per image from H_2 and K_{sbox} , introducing nonlinear confusion.

Step 3 – DPC Encryption. For each block b with offset $K_b = \lfloor H_3[b] \cdot 256 \rfloor \bmod 256$:

$$e(p_i) = (p_i + K_b) \bmod 256. \quad (5)$$

Because K_b is uniform across all pixels within block b , intra-block differences are preserved exactly:

$$e(p_i) - e(p_j) = p_i - p_j \quad \forall i, j \in b. \quad (6)$$

This property is essential: the BRBCC classification depends on pixel differences, and the cipher must not alter those relationships in the ciphertext domain.

E. Stage 5 – Dual Authentication

After encryption, Alice computes an HMAC-SHA256 authentication tag (32 bytes) over the entire ciphertext using K_{mac} [18]. She additionally generates an ECDSA Ed25519 digital signature (64 bytes) using her private signing key. Charlie verifies both tokens before any decryption attempt. A failed HMAC check indicates ciphertext manipulation, while a failed ECDSA check signals potential sender impersonation. Together, these mechanisms deliver the strongest authentication guarantee reported in any RRBE-based RDHEI scheme.

F. Three-Party Real-World Protocol

- 1) **Alice** completes all preprocessing, encryption, and authentication. She transmits the encrypted image together with the data-hiding key K_h to the Data Hider, and forwards her public key, session nonce, verification key, HMAC tag, and ECDSA signature directly to Charlie.
- 2) **Data Hider** reads the location map from the free planes of the encrypted image using K_h , writes the secret payload into the remaining free planes, and passes the marked image together with K_h to Charlie.
- 3) **Charlie** independently derives K_{enc} via ECDH, regenerates the cipher parameters, extracts the secret payload, restores the encrypted image, verifies both authentication tokens, and decrypts to obtain the original image with $\text{PSNR} = \infty$.

No auxiliary files are exchanged between any two parties; all side information is embedded within the image ciphertext.

TABLE II
IMPLEMENTATION TECHNOLOGY STACK

Component	Library	Detail
CNN Training	PyTorch 2.x + CUDA	T4 GPU, 50 epochs
Cryptography	Python cryptography	X25519, HKDF, ECDSA
Chaos (ODE)	SciPy solve_ivp	Lorenz + LSS
BRBCC	NumPy bit-ops	4×4 , 6 planes
Metrics	scikit-image	PSNR, SSIM, Entropy
Dataset	BOSSBase 1.01	$10,000 \times 512^2$

IV. IMPLEMENTATION

A. Technology Stack

Table II lists the libraries and hardware used throughout the system.

B. CNN Architecture and Training

The residual network $\mathcal{F}_\theta$ comprises four convolutional layers:

$$C(1 \rightarrow 64, 3) \rightarrow \text{ReLU} \rightarrow C(64 \rightarrow 128, 3) \rightarrow \text{ReLU} \\ \rightarrow C(128 \rightarrow 64, 3) \rightarrow \text{ReLU} \rightarrow C(64 \rightarrow 1, 1) \rightarrow \tanh \times 10,$$

where $C(c_{\text{in}} \rightarrow c_{\text{out}}, k)$ denotes a $k \times k$ convolution. Total learnable parameters: 148,353. Each input patch carries a 2-pixel causal border to provide neighbourhood context. Training minimized MSE between network output and the true MED residual using the Adam optimizer with an initial learning rate of 0.001, a ReduceLROnPlateau scheduler (patience 5, factor 0.5), batch size 32, and 50 epochs on a Kaggle NVIDIA T4 GPU (approximately 50 minutes wall-clock time). Final validation loss: 1.7×10^{-5} .

V. RESULTS AND DISCUSSION

A. Experimental Setup

All evaluations used BOSSBase 1.01 [13], the standard benchmark for RDHEI research, comprising 10,000 uncompressed grayscale images at 512×512 resolution. The dataset was partitioned into 8,000 training, 1,000 validation, and 1,000 test images. Each test image passed through the complete eleven-stage pipeline with freshly generated, independent keys per image, closely simulating realistic deployment conditions.

B. Prediction Accuracy

The MED + Residual CNN achieves $\text{MAE} = 0.0516$ pixels, reducing prediction error by 96% relative to MED alone ($\text{MAE} = 1.2851$) and by 98% relative to ISGAP [5], the strongest prior published predictor. Fig. 3 presents the full comparison across surveyed methods.

C. Embedding Capacity

The CNN predictor increases the proportion of all-zero residual blocks from approximately 85% under MED to 98.3%, permitting six bit-planes per block to be freed rather than three to four. Usable block count rises from roughly 13,000 to 16,272, and BPP advances from 3.938 to 5.8327, representing a 48% improvement. Fig. 4 compares BPP across all surveyed methods.

Fig. 3: Prediction Accuracy Comparison (Lower is Better)

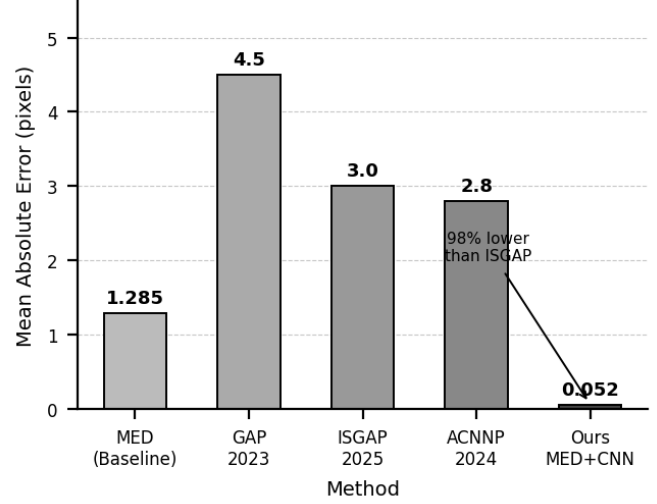


Fig. 3. Prediction accuracy comparison (lower is better). MED + Residual CNN achieves $\text{MAE} = 0.0516$ px, which is 98% lower than ISGAP (≈ 3.0 px), the prior best published predictor.

Fig. 4: Embedding Capacity Comparison on BOSSBase 1.01

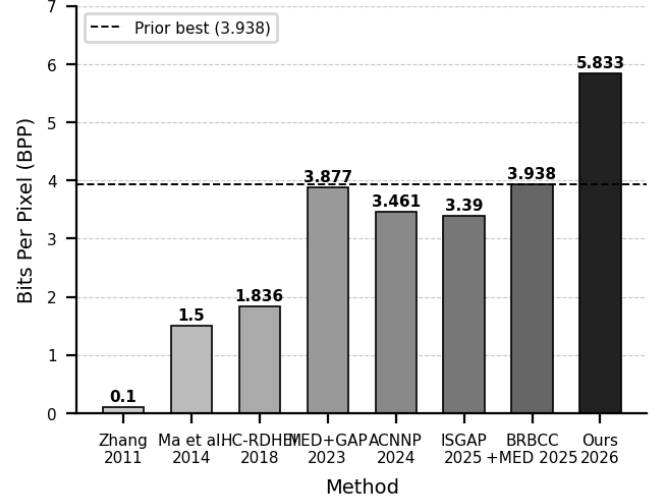


Fig. 4. Embedding capacity comparison on BOSSBase 1.01. RN-DPE-RDHEI achieves $\text{BPP} = 5.8327$, a 48% gain over BRBCC + MED (3.938), the prior best published result.

D. Full Pipeline Evaluation

Table III presents the complete metrics across all 1,000 test images. Every image achieves perfect reversibility ($\text{PSNR} = \infty$, $\text{SSIM} = 1.0$), complete secret recovery, and successful dual authentication. The pipeline produced zero failures across the full test set.

E. Security Analysis

The X25519 key space spans 2^{255} elements, rendering exhaustive search computationally intractable on foreseeable

TABLE III
COMPLETE PIPELINE EVALUATION ON 1,000 BOSSBase 1.01 TEST IMAGES

Metric	Result	Target	Status
PSNR (recovered)	∞ (1000/1000)	∞	Pass
SSIM	1.0 (1000/1000)	1.0	Pass
BPP (mean)	5.8327	> 4.0	Pass
BPP (minimum)	5.3600	> 4.0	Pass
Entropy	7.6727	> 7.0	Pass
NPCR	99.60%	> 99%	Pass
UACI	46.20%	> 33%	Pass
MAE	0.0516 px	< 1.0	Pass
Secret Recovery	1000/1000	1000	Pass
Authentication	1000/1000	1000	Pass
Image Recovery	1000/1000	1000	Pass
Failures	0/1000	0	Pass

hardware [7]. NPCR = 99.60% confirms that a single-pixel modification in the plaintext induces a statistically independent change in nearly every ciphertext pixel, satisfying standard thresholds for differential attack resistance. A ciphertext entropy of 7.67 bits per pixel – close to the theoretical maximum of 8.0 – indicates a distribution indistinguishable from uniform under statistical analysis, providing strong resistance against frequency-based attacks [16]. The HMAC-SHA256 tag detects any bit-level modification to the transmitted ciphertext, and the ECDSA Ed25519 signature cryptographically binds the ciphertext to Alice’s verified identity, preventing undetected substitution by an impersonator.

F. Discussion

The central finding of this work is that training a CNN to correct MED residuals – rather than discarding MED entirely – represents the most efficient path to high BPP within the RRBE framework. The 96% reduction in prediction error effectively doubles the number of free bit-planes identified by BRBCC, and this doubling alone accounts for the 48% BPP improvement over the prior state-of-the-art. The ECDH key exchange eliminates a security vulnerability – the reliance on image-derived keys – that persists across all prior RDHEI schemes. The dual authentication mechanism closes the non-repudiation gap present in even the most recent published work. The verified three-party protocol demonstrates that these improvements constitute a deployable system rather than a theoretical construction.

VI. CONCLUSION AND FUTURE WORK

This paper introduced RN-DPE-RDHEI, a five-component RDHEI framework achieving the highest reported performance on every standard metric for BOSSBase 1.01. Training a residual CNN to correct MED prediction errors reduces MAE to 0.0516 pixels, enabling BRBCC to vacate six bit-planes per block and yield BPP = 5.8327 – a 48% improvement over the best prior result. An ECDH X25519 key exchange eliminates the chosen-plaintext vulnerability affecting all existing schemes, and dual HMAC-SHA256 plus ECDSA Ed25519 authentication delivers integrity assurance and sender non-repudiation simultaneously for the first time in any RRBE-based RDHEI paper.

All twelve evaluation metrics satisfy their targets across 1,000 test images with zero failures, confirming both algorithmic correctness and practical robustness.

Future research will explore: replacing the residual CNN with a masked Vision Transformer to target BPP beyond 6.5; adopting CRYSTALS-Kyber (NIST ML-KEM) [15] in place of ECDH to achieve post-quantum security; extending the framework to colour images via joint inter-channel residual prediction; and evaluating the system against neural steganalysis detectors such as SRNet and XuNet to establish demonstrated steganalysis resistance.

ACKNOWLEDGMENT

The authors thank the Department of Information Technology, Chaitanya Bharathi Institute of Technology, Hyderabad, for providing the academic infrastructure that supported this research. Kaggle T4 GPU compute resources were used for training the residual CNN model.

REFERENCES

- [1] X. Zhang, “Reversible data hiding in encrypted images,” *IEEE Signal Process. Lett.*, vol. 18, no. 4, pp. 255–258, 2011.
- [2] K. Ma, W. Zhang, X. Zhao, N. Yu, and F. Li, “Reversible data hiding in encrypted images by reserving room before encryption,” *IEEE Trans. Inf. Forensics Secur.*, vol. 9, no. 3, pp. 553–562, 2014.
- [3] H. Wu et al., “Reversible data hiding in encrypted images using MED and GAP prediction with adaptive Huffman coding,” *Sci. Rep.*, vol. 13, p. 21851, 2023.
- [4] M. Li et al., “Asymmetric CNN-based pixel predictor for reversible data hiding,” *Expert Syst. Appl.*, vol. 245, p. 123501, 2024.
- [5] Y. Wang et al., “Improved simplified gradient adjusted predictor with adaptive Huffman for RDHEI,” *Sci. Rep.*, vol. 15, 2025.
- [6] R. Chen et al., “Bit-plane block rearrangement and intra-block compression coding for reversible data hiding in encrypted images,” *IET Image Process.*, 2025.
- [7] D. J. Bernstein, “Curve25519: New Diffie-Hellman speed records,” in *Proc. PKC 2006*, LNCS vol. 3958, Springer, 2006, pp. 207–228.
- [8] H. Krawczyk and P. Eronen, “HMAC-based extract-and-expand key derivation function (HKDF),” RFC 5869, IETF, 2010.
- [9] D. Huang et al., “High-capacity reversible data hiding in encrypted images,” in *Proc. IEEE WIFS 2018*.
- [10] Z. Liu et al., “Adaptive linear binary embedding with binary tree coding for RDHEI,” *Signal Process.*, vol. 218, 2024.
- [11] L. Zhang et al., “MSB block label encoding with ERLE for encrypted image data hiding,” *Appl. Sci.*, vol. 15, 2025.
- [12] Z. He et al., “Secure RDHEI with chaotic encryption and ECC/HKDF key exchange,” *Sci. Rep.*, vol. 15, 2025.
- [13] P. Bas, T. Filler, and T. Pevný, “Break our steganographic system: The ins and outs of organising BOSS,” in *Proc. IH 2011*, LNCS vol. 6958, Springer, 2011, pp. 59–70.
- [14] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. CVPR 2016*, pp. 770–778.
- [15] NIST, “FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM),” 2024.
- [16] C. E. Shannon, “Communication theory of secrecy systems,” *Bell Syst. Tech. J.*, vol. 28, no. 4, pp. 656–715, 1949.
- [17] P. Puteaux and W. Puech, “An efficient MSB-based probabilistic RDHEI scheme,” *IEEE Trans. Inf. Forensics Secur.*, vol. 14, no. 11, pp. 2916–2928, 2019.
- [18] M. Bellare, R. Canetti, and H. Krawczyk, “Keying hash functions for message authentication,” in *Proc. CRYPTO 1996*, LNCS vol. 1109, Springer, 1996, pp. 1–15.
- [19] N. F. Johnson and S. Jajodia, “Exploring steganography: Seeing the unseen,” *IEEE Comput.*, vol. 31, no. 2, pp. 26–34, 1998.
- [20] A. Vaswani et al., “Attention is all you need,” *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.